DQE

Data Quality Engine

Automated PMS–Duetto discrepancy detection and root-cause analysis

For DVA-supported integrations · Internal Use — Duetto Employees   Version 1.0 · June 2026

What is the Data Quality Engine?

The **Data Quality Engine (DQE)** is a Duetto-built web application that automates the identification and root-cause analysis of discrepancies between a hotel's **Property Management System (PMS)** and the data held inside **Duetto**. It is designed for use by **Duetto Quality Analysts** with use cases for **Duetto Support Agents**.

Rather than manually comparing exports line by line, the DQE ingests multiple source files, applies a library of classification rules, computes accuracy metrics, and generates a structured report — in seconds.

DVA-Supported Integrations

The DQE is built around the **Daily Variance Analysis (DVA)** workbook, the standard output of Duetto's PMS integration layer. Any integration that produces a DVA file is compatible with the engine. The following integrations have been tested and validated:

PMS Platform	Integration Type	Notes
Opera / Opera Cloud	XML / API	Primary target; Arrival Details Report supported
Infor HMS	File-based	Validated; DVA export required
Any DVA-producing PMS	—	Compatible if standard DVA columns are present

*The DVA workbook is the only **required** input. All other source files are optional and progressively unlock additional analysis capabilities.*

Source File Inputs

The DQE accepts up to five source files through a drag-and-drop upload interface. Each file unlocks an additional layer of diagnostic capability:

File	Format	Required	Capability Unlocked
DVA (Daily Variance Analysis)	.xlsx	✓ Yes	Core rooms & revenue comparison; hotel name extraction
Bookings Stay Date Report	.tsv / .csv	Optional	Reservation-level root-cause analysis; missing/phantom booking detection
Blocks Stay Date Report	.tsv / .csv	Optional	Block reservation context for room discrepancies
Flat Folio Report	.tsv / .csv	Optional	Folio-level revenue analysis; no-show fees, adjustments, refunds

File	Format	Required	Capability Unlocked
Arrival Details Report (Opera)	.xml / .pdf / .txt	Optional	PMS-to-Duetto cross-reference; sync-gap detection & rate variance

Analysis Engine & Discrepancy Classification

For every stay date in scope, the DQE computes room and revenue accuracy against PMS-reported values from the DVA. Failing dates are classified using a library of named diagnostic codes that identify the most likely root cause:

Code	Category	Description
HO-O-10	Historic – Rooms Over	Stale RESERVED-status bookings on actualized dates
HO-O-11	Historic – Rooms Over	Phantom active bookings — cancellations not synced
HO-O-13	Historic – Rooms Over	Orphaned share reservations in Duetto
HO-U-05	Historic – Rooms Under	Leg-perm / booking leg configuration issue
HO-U-06	Historic – Rooms Under	Missing bookings — integration publisher gap
FO-O-45	Future – Rooms Over	Phantom future bookings — cancellation not received
FO-O-46	Future – Rooms Over	Stale reservations — resync required
FO-U-40	Future – Rooms Under	Missing future bookings — integration setup issue
HR-U-14	Historic – Revenue Under	No-show fees captured in folio but not by Duetto
HR-U-21	Historic – Revenue Under	Active bookings carry $0 rate in Duetto
HR-O-24	Historic – Revenue Over	Folio refund/cancellation fee — PMS negative adjustment
FR-U-51	Future – Revenue Under	Future bookings with $0 or suppressed rate in Duetto
FR-O-53	Future – Revenue Over	Future folio credit — Duetto does not capture adjustments
SYNC-GAP	PMS Integration Gap	Opera Arrivals & Duetto agree, but DVA PMS commit is lower — OHIP sync feed incomplete

Each discrepancy record includes the diagnostic code, a plain-English explanation, and a list of contributing bookings (confirmation numbers, statuses, rates) to support immediate action.

Accuracy Metrics

The DQE calculates two headline accuracy percentages across the scoped date range:

- **Room Accuracy** — percentage of stay dates where Duetto's committed room count matches the PMS exactly (zero-tolerance threshold).
- **Revenue Accuracy** — percentage of stay dates where Duetto's committed revenue is within the acceptable variance of the PMS value.

Dates are scoped automatically to the union of dates present in the uploaded source files — allowing short-range targeted analysis without reprocessing the full calendar.

Arrival Details Report — Stay Date Reconciliation

When an **Opera Arrival Details Report** is uploaded alongside the Duetto Bookings Stay Date Report, the DQE reconciles three independent sources of room count for the stay date(s) in scope. The report is accepted in **.xml, .pdf, or .txt (tab-delimited)** format — the engine auto-detects the format and parses accordingly.

The reconciliation surfaces a three-way comparison:

- **Opera Arrivals (PMS)** — the true reservation count exported directly from Opera.
- **Duetto Bookings** — the reservation count held in Duetto, matched by Opera confirmation number (ALTERNATE_SOURCE_ID).
- **DVA PMS Commit** — the PMS room commit figure recorded in the Daily Variance Analysis, fed by the OHIP sync.

Sync-gap detection. The most valuable pattern this exposes is when Opera Arrivals and Duetto Bookings *agree* (e.g. both 71) but the DVA PMS commit is lower (e.g. 42). This proves the discrepancy is not a Duetto accuracy problem — it is an incomplete OHIP sync feed under-reporting to the DVA. The affected stay date is re-classified as **SYNC-GAP** (amber) rather than a Duetto "Over" failure, and the exact room gap is called out on-screen and in the export.

- **Rate mismatches** — flags reservations where the Opera effective rate differs from the Duetto rate by more than $1.00, surfacing ADR variance root causes.

Results appear in a dedicated **Stay Date Reconciliation** panel and the reframed accuracy tile in the on-screen report, and in a dedicated **Arrivals Reconciliation** sheet in the Excel export (three-source comparison, plain-English summary, and a full rate-mismatch detail table). The original source file is also uploaded to the Monday.com board for audit purposes.

Report Outputs

Output	Description
On-screen Report	Interactive accuracy dashboard with discrepancy table, folio analysis, arrival cross-reference, and remediation recommendations — rendered immediately in the browser.
Excel Workbook (.xlsx)	Structured export with a Discrepancy Report sheet (rooms & revenue, root-cause codes, recommendations) and — when an Arrival Details Report is supplied — an Arrivals Reconciliation sheet showing the three-source room comparison, sync-gap summary, and rate-mismatch detail. Branded with Duetto colours.
Monday.com Board Item	Automatic submission to the DQE Feedback board — includes hotel metadata, accuracy scores, failing dates, all source files, and a notes field for the submitting employee.

Monday.com Integration

After running an analysis, results can be submitted directly to the shared **DQE Feedback board on Monday.com** via a single-click modal. The following data is captured on each board item:

- Hotel Name and Hotel ID (auto-filled from the DVA filename)
- Analysis date, stay date range, rooms accuracy %, revenue accuracy %
- Failing dates summary
- Files used (list of all uploaded source file names)
- Submitted By — workspace member dropdown populated from Monday.com
- Free-text Feedback / Notes field

- All source files attached in dedicated columns (DVA, Bookings, Blocks, Folio, Arrivals, Excel Report)

The board provides a centralised audit trail of all DQE submissions across the team, with attached files retained for review and escalation.

Board link: https://duettoresearch.monday.com/boards/18419945182

Recommendations Engine

Based on the frequency and type of diagnostic codes identified, the DQE automatically generates a prioritised set of recommended remediation actions for Duetto employees to review and act on. Examples include:

- Request a historical reservation resync to clear stale RESERVED-status bookings.
- Review integration XML publisher settings for missing booking codes.
- Investigate leg-perm configuration with the integration manager.
- Manually cancel phantom future bookings; verify the integration receives delete messages.
- Review folio-level integration options where no-show fee revenue is systematically absent.

Recommendations are included in both the on-screen report and the Excel export, and grouped by diagnostic code to support efficient action planning by Duetto employees.

Running the DQE

The DQE runs as a local Flask web application:

Step	Action
1	Open Terminal and navigate to the DQE folder
2	Run: python app.py
3	Open a browser and go to: http://localhost:5055
4	Upload source files using the drag-and-drop zones
5	Click Run Analysis and review the on-screen report
6	Download the Excel report and/or Submit to Monday.com

Requires Python 3.9+ with dependencies installed via pip. No internet connection is needed for the analysis itself; Monday.com submission requires network access.

Key Benefits

Speed

Reduces manual DVA review from hours to seconds. A full analysis across a 365-day date range completes in under 10 seconds.

Consistency

Every discrepancy is classified against the same ruleset — eliminating analyst-to-analyst variance in root-cause attribution.

Depth

Booking-level, block-level, folio-level, and PMS-level evidence is surfaced automatically — no manual cross-referencing required.

Traceability

Every submission creates an auditable Monday.com record with source files attached, supporting QA, escalation, and hand-off workflows.

Scalability

Distributable as a local app — any Duetto employee can run it independently without a shared server or cloud dependency.

The Data Quality Engine is an internal Duetto tool. For questions, issues, or feature requests, contact the DQE team via the Monday.com DQE Feedback board or Slack.